Can Machine Learning Predict If Your TikTok Will Be a Hit or a Flop ?

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This study aims to predict the number of likes a TikTok video will get based on its characteristics. Using a dataset of 19 382 TikTok video metadata, regression models were tested including Linear regression, LASSO regression, and a Neural Network. Model performance was evaluated using RMSE. Results show that the Neural Network achieved the lowest test RMSE (44 532), outperforming the linear and LASSO models.

1. Introduction

Predicting the number of likes a TikTok video will receive is a valuable task in social media analytics. It can guide content creation, advertising strategies, and platform optimization.

My goal is to predict the number of likes of a TikTok video based on its various characteristics and user metadata.

Because I want to predict a continuous variable (number of likes), it is a **regression goal**.

2. Data

The dataset was made in 2023 by collecting a wealth of information on many TikTok videos, and the **number of likes (N = 19 084, see Figure 1).**

Each TikTok video is described by **12 features** (verified user or not, number of views, comments, shares, video duration, text transcription, etc.).

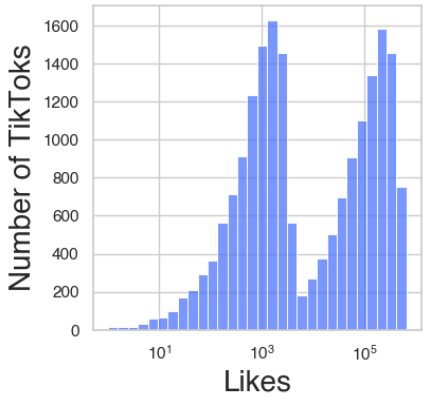


Figure 1 – Distribution of the dependent variable 'Likes'

3. Method

I created **25 new variables :**

- Log transformations & simple ratios (shares per view, comments per view, ...)
- Encoded version of categorical variables (verified user, banned user or not, ...)
- Duration based features (views per second, comments per second, ...)

I then **standardized** the features. I separated the data into training set (75%) and testing set (25%). I then built and evaluated three **regression models** :

- **Linear regression**
- **LASSO regression** (5-fold cross-validated for 35 fits, $1/\lambda \in \{1, 10, 100, 1000, 10\,000, 100\,000, 1\,000\,000\}$)
- **Neural Network** (5-fold cross-validated for 720 fits and several hyper-parameters, **see Table A1**)

I aim to minimize large prediction errors. I use the **RMSE** which penalizes larger deviations.

4. Results

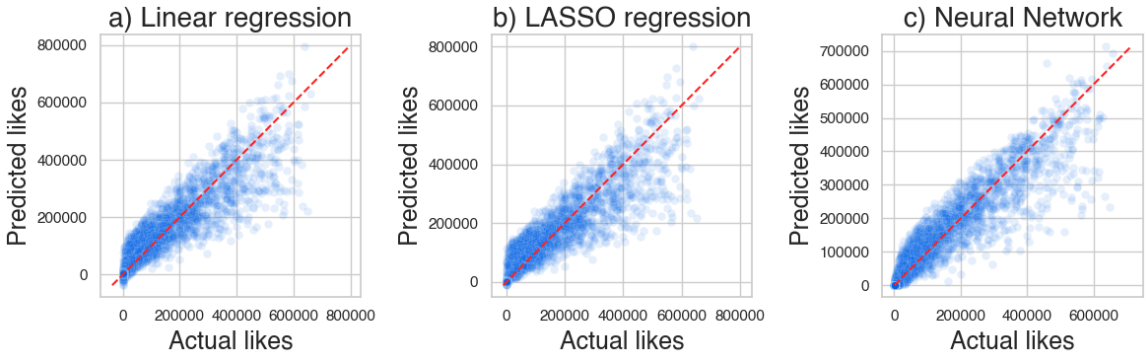


Figure 2 – Scatterplot of predicted versus actual values of likes for : **a) Linear regression, b) LASSO regression, and c) Neural Network**

Model	Train RMSE	Test RMSE
Linear regression	49 435	49 926
LASSO regression	49 881	50 409
Neural Network	44 611	44 532

Table 1 – Summary of train RMSE and test RMSE

4.1. Linear regression

The RMSE of this model is **49 435** on the training set and increase slightly to **49 926** on the testing set **(see Figure 2.a & Table 1)**. This small gap suggests low variance. The value of test RMSE indicates that the model might suffer from bias.

4.2. LASSO regression

The RMSE of this model is **49 881** on the training set and increase slightly to **50 409** on the testing set **(optimized 1/λ = 1000, see Figure 2.b & Table 1)**. This small gap suggests quite low variance. The value of test RMSE indicates that the model might suffer from bias.

4.3. Neural Network

The RMSE of this model is **44 611** on the training set and decrease slightly to **44 532** on the testing set **(see Figure 2.c, Table 1 & Table A1** for optimized hyper-parameters**)**. This small gap suggests very low variance. The value of test RMSE indicates that the model might suffer from bias. The error curve shows that the model needs some iterations to stabilize because of the quite large amount of data. The model continues to reduce training error slowly while test error plateaus around 200 iterations, indicating a potential bias **(see Figure 3)**.

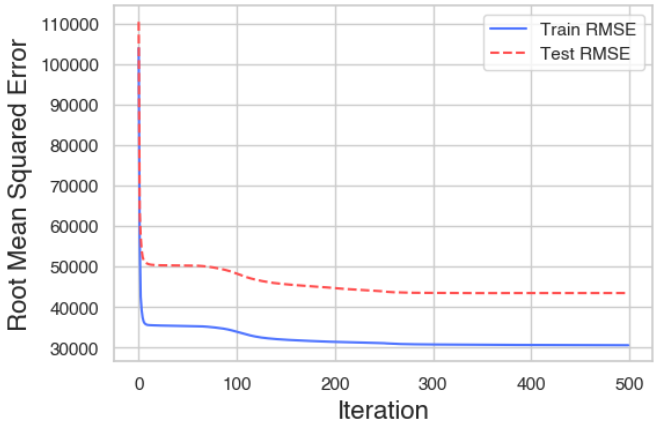


Figure 3 – Error curve of the Neural Network

5. Conclusion

The best model is the **Neural Network**, with a **test RMSE of 44 532**. The model predicts likes with an average error about 44k likes, indicating a **good accuracy**. Thus, user metadata and TikTok videos' characteristics do provide some nice predictive signals to predict likes.

Hyper-parameter	Tested values	Optimized values
Hidden layer sizes	[1], [2], [2,2], [4,2], [8,4], [16,8]	[8,4]
Activation function	ReLU, Tanh, Logistic	ReLU
Solver	adam, sgd	adam
Initial learning rare	0.001, 0.01	0.01
Batch sizes	32, 64	32

Table A1 – Hyper-parameters tested and optimized in 5-fold cross-validation